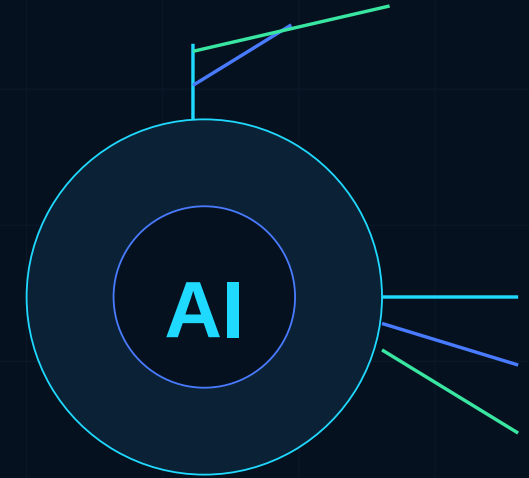


AI-POWERED SECURITY
OPERATIONS

AI Cyber Threat Intelligence System

From login signal to automated defense.



JUDGES' PRESENTATION • 2026

Secure Login • Live SOC • Threat AI • SOAR

Security teams see events. They need decisions.

Authentication signals are often scattered across tools, slowing detection and response.

FRAGMENTED VISIBILITY

Login activity, threat intelligence, incidents and reports live in separate workflows.

ALERT OVERLOAD

Raw events create noise when they are not correlated, scored and explained.

SLOW RESPONSE

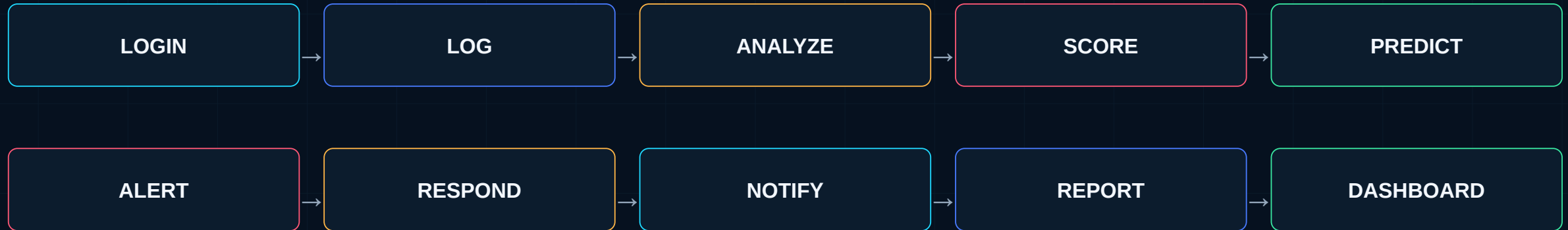
Analysts lose valuable time moving from detection to investigation and defensive action.

THE GAP

A continuous, explainable security loop—from authentication to response.

One defensive pipeline. One operational view.

Sentinel AI turns every authentication event into actionable SOC intelligence.



THE VALUE

Faster understanding • Explainable risk • Consistent response • Auditable evidence

A threat story judges can see end-to-end

The demonstration begins with a normal user action and ends with an SOC response.

0
1

Secure login

Successful and failed attempts are captured with user, IP and device context.

0
2

Detection

>5 failures for the same user and IP in 5 minutes triggers brute-force logic.

0
3

AI risk

Behavior is scored from 0–100 and classified LOW, MEDIUM or HIGH.

0
4

SOC response

Alert, attack path, incident, notification and report become visible.

Built for the analyst's next decision

A responsive command center unifies real-time monitoring, intelligence and response.

LIVE MONITORING

Authentication timeline, search, filters and status signals

THREAT INTELLIGENCE

Suspicious IPs, threat types and contextual risk

AI DETECTION

Risk score, confidence and explainable behavior analysis

ATTACK GRAPH

Likely progression from entry point to sensitive asset

SOAR RESPONSE

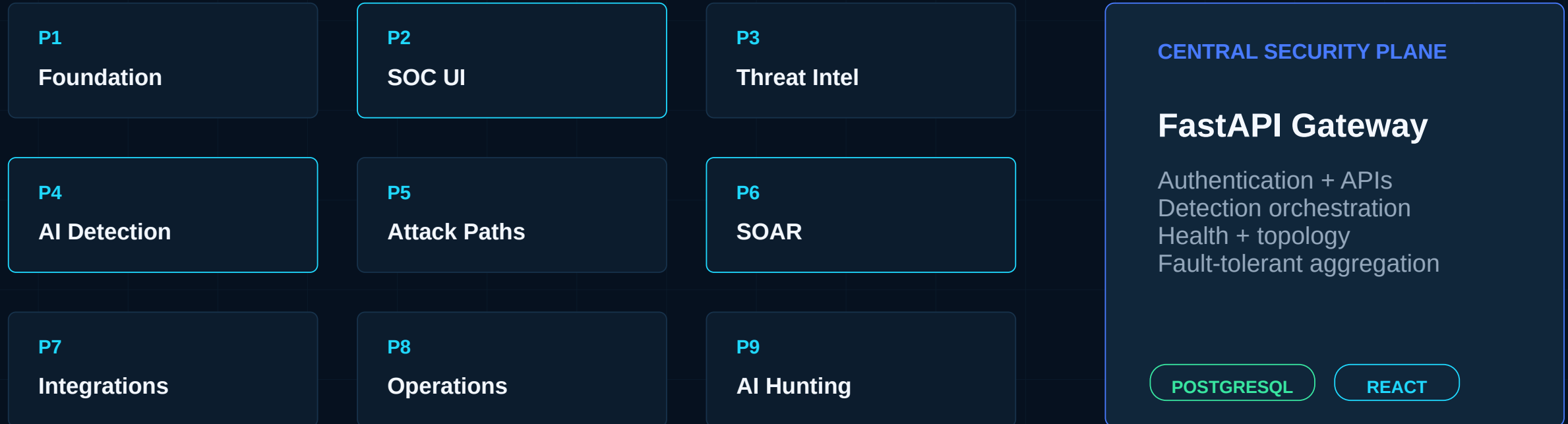
Incident workflow from detection through resolution

REPORTING

Daily, threat and incident reports with export-ready summaries

Modular by design. Integrated by a central security plane.

Nine independently runnable phases remain intact while the Central API coordinates data and operations.



Risk that analysts can understand—and challenge

Deterministic controls provide a reliable baseline; Phase 4 extends the pipeline with AI detection.

NORMAL ACTIVITY

10 / 100

Known behavior and
successful authentication

SUSPICIOUS PATTERN

35–55

Repeated failures or
abnormal login behavior

BRUTE FORCE

90 / 100

Threshold crossed for the
same account and source

EXPLAINABILITY

**Every score keeps its
evidence trail:**

- username behavior
- source IP activity
- attempt frequency
- event window
- detection reason

Human-readable reasoning remains visible beside the score.

The security platform protects its own trust boundary

Controls are implemented in the central authentication and persistence layer.

PBKDF2

Password hashing with 600,000 iterations

TOKEN SAFETY

Hashed bearer tokens with expiry and revocation

SECRET HYGIENE

Credentials and keys supplied through environment variables

SAFE MIGRATION

Additive initialization preserves existing users and data

AUDITABILITY

Success, failure, user agent, IP and reason are recorded

DEFENSIVE SCOPE

Automated actions prepare and record defensive response

More than a dashboard: a closed-loop SOC platform

END-TO-END

A single flow connects authentication, logs, intelligence, AI, prediction, response, notifications and reports.

MODULAR

Nine phases keep clear ownership and can evolve independently behind a central integration layer.

OPERATIONAL

Health checks, fault-tolerant aggregation, auditable evidence and safe response make the system demo-ready and extensible.

DETECT

UNDERSTAND

PREDICT

RESPOND

Validated as a working system—not a static concept

The project includes executable services, persistence, APIs and a connected frontend.

FRONTEND

BUILD

Production build verifies the React command center and report experience.

BACKEND

COMPILE

Python compile checks protect API, model and service integrity.

INTEGRATION

TEST

Isolated tests exercise login logging, brute-force detection, notifications and report generation.

OPERATIONS

RUN

Docker Compose, health endpoints and environment-based configuration support repeatable startup.

DEMO SUCCESS CRITERIA

Failed login → notification | 6 attempts → high alert | report → persisted summary

A clear path from prototype to enterprise deployment

NOW

Harden

RBAC • WebSocket streaming •
Geo/IP enrichment • expanded
automated tests

NEXT

Integrate

SIEM feeds • identity providers •
threat exchanges • model
observability

SCALE

Enterprise

Multi-tenancy • HA deployment •
compliance packs • analyst
collaboration

THE
MISSION

Turn every security signal into a confident response.

Sentinel AI connects detection, understanding, prediction and response in one explainable SOC workflow.

LIVE DEMO

LOGIN



DETECT



RESPOND

DETECT • UNDERSTAND • PREDICT • RESPOND

Thank you / Questions